

SIMPLE IMPACT STUDIES — SAMPLE DRAFT

Greenville County Sample

Traffic Impact Study

Prepared September 2, 2026 · 34.8290, -82.3510



Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Study evaluates the proposed Greenville County Sample within Greenville-Spartanburg (SC), South Carolina (Greenville County), in accordance with SCDOT's Access and Roadside Management Standards (ARMS, 2008 edition, updated July 8, 2025) Chapter 6 and Traffic Engineering Guideline TG-21. In South Carolina the TIS is an element of the SCDOT encroachment-permit process: the study scope must be established with the District Traffic Engineer (DTE) before analysis begins, and the DTE makes the final mitigation determination.

7	680	1	1
INTERSECTIONS	PM PEAK TRIPS	AT LOS E/F	LOS DROPS

1.0 INTRODUCTION

Project name	Greenville County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	34.8290°, -82.3510°
Build-out year	2027
Controlling jurisdiction	Greenville County
MPO / study	GPATS (Greenville-Pickens Area Transportation Study)
Programmed projects source	SCDOT FFY 2024–2033 STIP, as amended (e-STIP)
Submittal channel	SCDOT EPPS (Encroachment Permit Processing System) via the District Permit Engineer

1.1 TIS Applicability

Greenville County Land Development Regulations Article 9 (April 2018): TIS on county roads at 100+ peak-hour trips (50 in unzoned areas; expansions adding 25%+ peak-hour trips), default study area of the 3 highest-volume intersections within 1/2 mile (3/4 mile unzoned), with a fee-in-lieu mechanism where right-of-way cannot be obtained. State roads follow ARMS with the SCDOT Traffic Requirements Form filed with the County. The County UDO Sec. 22.8 (adopted Dec 2024) carries the 100 peak-hour-trip trigger forward.

The proposed development's estimated peak-hour generation of 680 trips meets or exceeds the ARMS 100 peak-hour-trip threshold; a TIS is required for the encroachment permit.

2.0 PROJECT TRAFFIC

ARMS Chapter 6 calls for trip generation per the latest edition of the ITE Trip Generation Manual, with internal-capture, pass-by, and transit reductions justified to the DTE. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal.

Daily rate applied	80.00 trips per ksf × 85 ksf
AM peak rate applied	3.20 trips per ksf × 85 ksf
PM peak rate applied	8.00 trips per ksf × 85 ksf
Rate source	SANDAG 2002 “(Not So) Brief Guide of Vehicular Traffic Generation Rates for the San Diego Region” — Community Shopping Center (daily 80/ksf; PM pass-by 30%)

Daily trips (gross)	6,800
AM peak hour	166 in / 106 out
PM peak hour	326 in / 354 out
Counts standard	AM and PM weekday peaks minimum; counts ≤ 12 months old, school in session, seasonally adjusted

2.1 Trip Distribution — Gravity Model

The trip distribution uses the gravity model ($\text{term} = M / (d^1 \cdot d_{\text{site}})$); mass basis: intersection through-volume ($\text{AADT} \times \text{K-factor}$) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

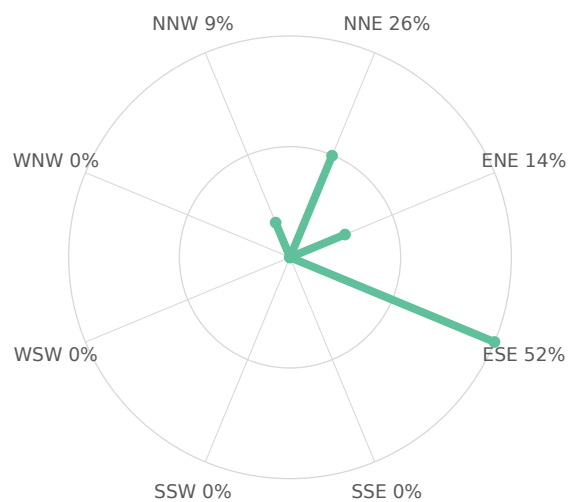
Directional sector		Share of project trips			
Northeast (NNE + ENE)		40%			
Southeast (ESE + SSE)		52%			
Southwest (SSW + WSW)		0%			
Northwest (WNW + NNW)		9%			

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Laurens Road & Henderson Road	ESE	0.47	3,339	0.3	34.52%
Near Laurens Road	ESE	0.27	1,535	0.2	17.15%
Laurens Road & Woodruff Road	ENE	0.19	1,206	0.1	13.89%
Laurens Road & Haywood Road	NNE	0.27	1,206	0.1	13.47%
Haywood Road & Halton Road	NNE	0.55	1,206	0.1	12.21%
Laurens Road & Industrial Drive	NNW	0.41	432	0.0	4.59%
Parkins Mill Road & Laurens Road	NNW	0.69	432	0.0	4.18%

7 study-area zone(s). Screening-grade distribution; not a calibrated regional model.



Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

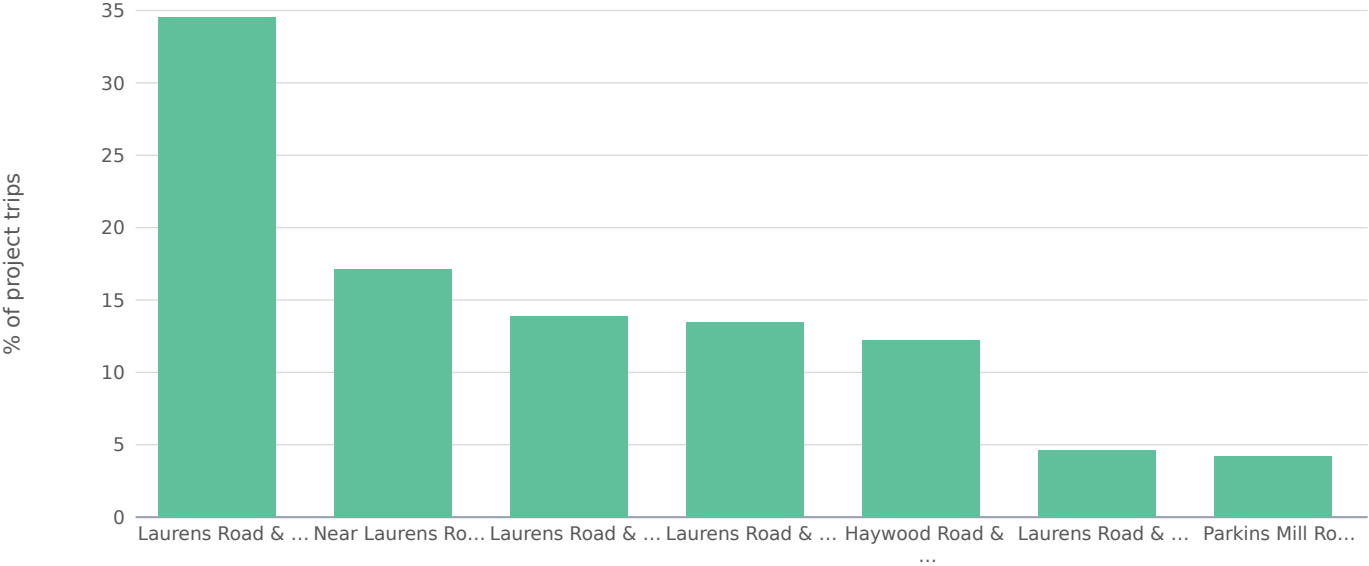
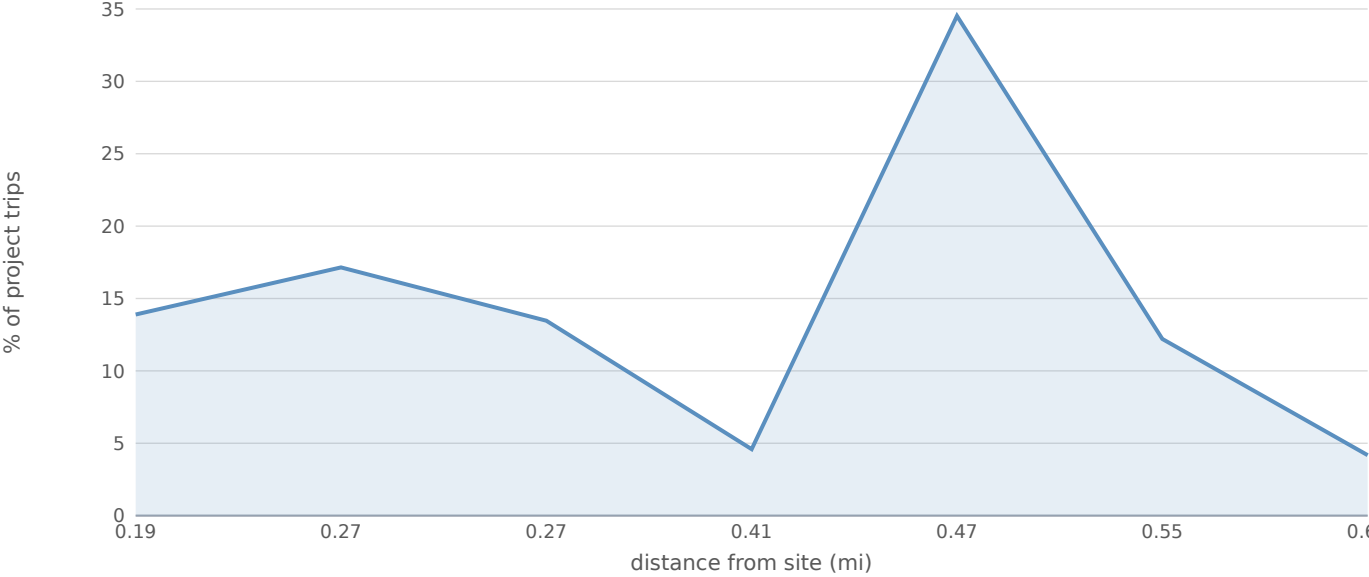


Figure — Trip Share vs. Distance from Site (Gravity Decay)



2.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Laurens Road & Woodruff Road	ENE	0.19	58	147	17.6%
Near Laurens Road	ESE	0.27	118	223	26.8%
Laurens Road & Haywood Road	NNE	0.27	36	53	6.4%
Laurens Road & Industrial Drive	NNW	0.41	20	38	4.6%
Laurens Road & Henderson Road	ESE	0.47	118	223	26.8%
Haywood Road & Halton Road	NNE	0.55	58	111	13.3%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Parkins Mill Road & Laurens Road	NNW	0.69	20	38	4.6%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

3.0 TRAFFIC VOLUME DEVELOPMENT AND CAPACITY ANALYSIS

Scenarios per ARMS: existing year, build-out no-build, and build-out build. Capacity analysis computes signalized control delay by the openly-published Webster/Akcelik method with LOS reported for all approaches and movements; coordinated signal systems are analyzed as a system.

3.1 TG-21 Level of Service Standard — LOS C

Per SCDOT Traffic Engineering Guideline TG-21 (Mitigation of Traffic Impacts), the acceptable level of service for the design (peak) hour is LOS C or better for all roadway types statewide, applied in lieu of locally preferred thresholds. Where the baseline already operates at or below LOS C, the baseline LOS must be maintained or improved; where the baseline is LOS F in a congested urban area, mitigation is at the DTE's determination. Note: this is a stricter standard than the LOS D convention applied in several neighboring states.

No.	Intersection	Exist.	No-Build	Build LOS/Delay	TG-21 (LOS C) check
1	Laurens Road & Woodruff Road	C	C	C / 27.4s	meets LOS C
2	Near Laurens Road	C	C	D / 53.7s	TG-21 MITIGATION INDICATED
3	Laurens Road & Haywood Road	C	C	C / 25.2s	meets LOS C
4	Laurens Road & Industrial Drive	B	B	B / 16.2s	meets LOS C
5	Laurens Road & Henderson Road	F	F	F / 300.0s	baseline maintained
6	Haywood Road & Halton Road	C	C	C / 26.5s	meets LOS C
7	Parkins Mill Road & Laurens Road	B	B	B / 16.2s	meets LOS C

4.0 FINDINGS, ACCESS MANAGEMENT, AND SIGNAL WARRANTS

1 intersection project a LOS drop and 1 operate at LOS E or F under Build conditions; mitigation alternatives (turn lanes with storage per ARMS Ch. 5 and the SCDOT Roadway Design Manual Ch. 9, signalization, or operational improvements) are developed with the DTE, who makes the final determination.

Access management: ARMS requires demonstration that the fewest driveways necessary serve the site, with sight-distance verification at each access point. Signal proposals require a signal warrant analysis per the MUTCD. Both are site-plan-dependent and are prepared at submittal.

5.0 ARMS TECHNICAL COMPLETENESS CHECKLIST

SCDOT returns incomplete studies unreviewed. The ARMS completeness checklist is tracked below against this screening draft; unmet items are produced at formal submittal.

Checklist item	Status in this draft
SC PE stamp and signature	At submittal (screening draft is unsealed)
Introduction + executive summary	Included
Existing conditions (classification, peaks, LOS, volumes)	Included (screening level)
Trip generation / distribution / LOS analysis	Included (public-data rates, tagged)

Checklist item	Status in this draft
Crash data / speeds / sight distance	At submittal (site-specific)
Mitigation w/ turn-lane storage lengths	Indicated per TG-21 table; design at submittal
Figures (vicinity, site plan, volumes, lane configs, LOS)	Volume/LOS figures included; site-plan figures at submittal
Tables (trip gen, LOS existing/background/build/mitigated)	Included (screening level)
Technical appendix (capacity reports, warrants)	Capacity worksheets appended; Synchro/warrants at submittal

This TIS is to be prepared under the direct charge of and sealed by a registered South Carolina Professional Engineer with expertise in traffic engineering (ARMS), with seal, signature across the seal face, and date per S.C. Code §40-22-270 and Regs. Ch. 49, and the firm's Certificate of Authorization. Electronic sealing is accepted in current practice.

SCREENING DISCLOSURE: this document is a screening-level draft prepared before DTE scoping. Scope, growth rates, background developments, and mitigation are established with the District Traffic Engineer; a submittal-grade analysis supersedes this draft.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Near Laurens Road	0.27	0.6	223	26.8%
Laurens Road & Henderson Road	0.47	1.1	223	26.8%
Laurens Road & Woodruff Road	0.19	0.5	147	17.6%
Haywood Road & Halton Road	0.55	1.3	111	13.3%
Laurens Road & Haywood Road	0.27	0.6	53	6.4%
Laurens Road & Industrial Drive	0.41	1.0	38	4.6%
Parkins Mill Road & Laurens Road	0.69	1.7	38	4.6%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 91% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 9% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.38. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	0.60	431	0.38
Principal arterial	0.93	282	0.30
Minor arterial	0.30	58	0.18

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c , delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 12 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: $\sum \text{entering} = \sum \text{leaving}$ at 267 network junctions, max

imbalance 0.00 veh. 7 study intersection(s) carry path-derived movements; 0 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

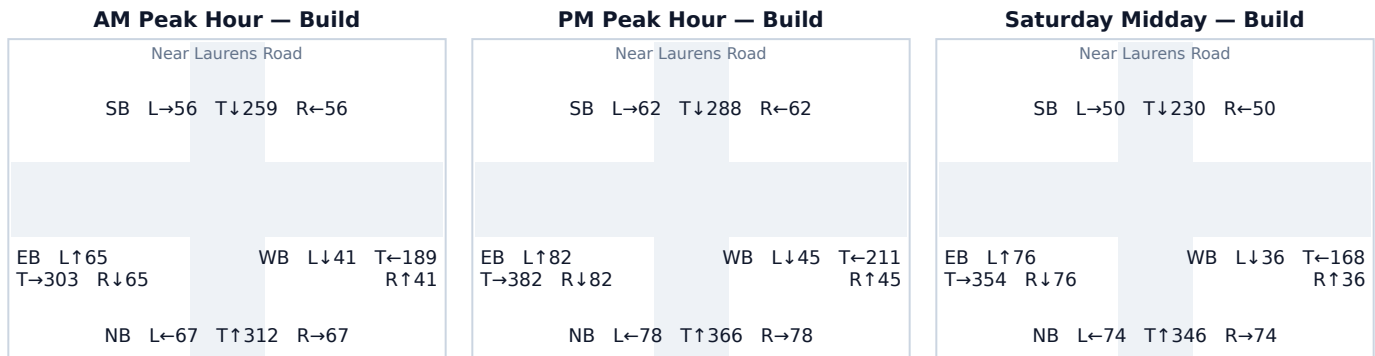
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Near Laurens Road

Signal ID	greenville-spartanburg-966
Location	NW Greenville-Spartanburg (SC) · 0.27 mi from site
Added PM peak trips	223
Existing / No-Build (PM)	LOS C · 34.2 s/veh · v/c 0.87
Build (PM)	LOS D · 53.7 s/veh · v/c 0.99
Δ control delay	+19.5 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	415	107	522	0.51 → 0.64	20.0 → 23.1	C → C	417
SB	412	0	412	0.51 → 0.51	19.9 → 19.9	B → B	303
EB	430	116	546	0.53 → 0.67	20.4 → 24.0	C → C	444
WB	301	0	301	0.37 → 0.37	17.6 → 17.6	B → B	205

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Right	116	52%
NB Through	107	48%

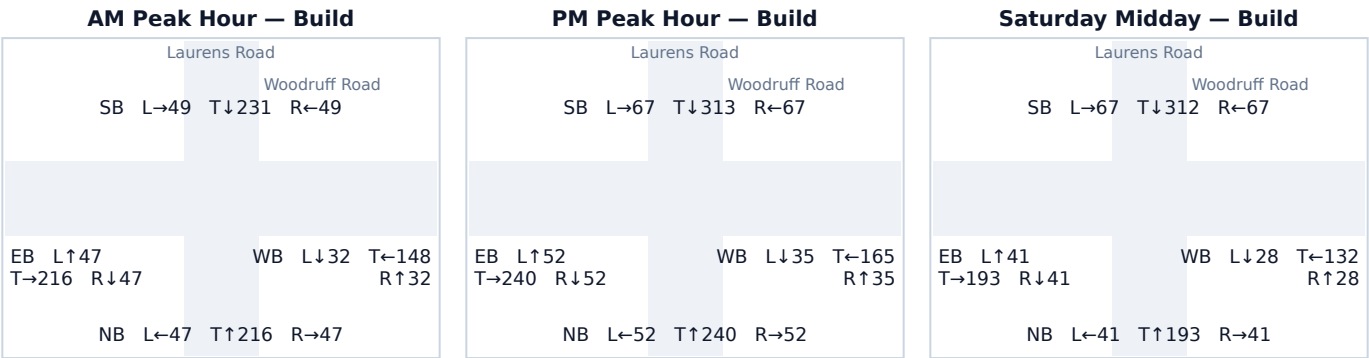
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Laurens Road & Woodruff Road

Signal ID	greenville-spartanburg-1196
Location	NW Greenville-Spartanburg (SC) · 0.19 mi from site
Added PM peak trips	147
Existing / No-Build (PM)	LOS C · 24.2 s/veh · v/c 0.68
Build (PM)	LOS C · 27.4 s/veh · v/c 0.76
Δ control delay	+3.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	344	0	344	0.42 → 0.42	18.5 → 18.5	B → B	241
SB	301	147	447	0.37 → 0.55	17.6 → 20.8	B → C	338
EB	344	0	344	0.43 → 0.43	18.5 → 18.5	B → B	242
WB	235	0	235	0.29 → 0.29	16.6 → 16.6	B → B	154

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	116	79%
SB Left	31	21%

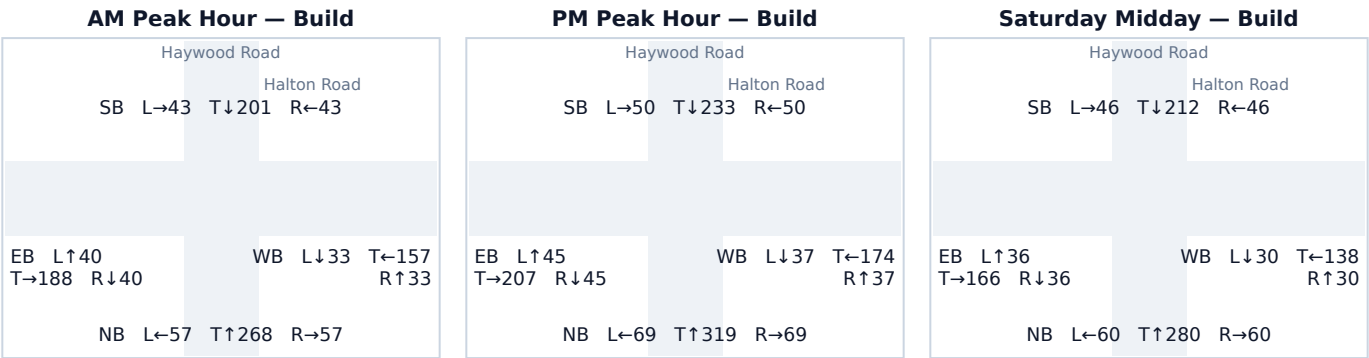
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A.3 Haywood Road & Halton Road

Signal ID	greenville-spartanburg-418
Location	NW Greenville-Spartanburg (SC) · 0.55 mi from site 111
Added PM peak trips	
Existing / No-Build (PM)	LOS C · 24.2 s/veh · v/c 0.68
Build (PM)	LOS C · 26.5 s/veh · v/c 0.74
Δ control delay	+2.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	399	58	457	0.49 → 0.56	19.6 → 21.1	B → C	347
SB	280	53	333	0.35 → 0.41	17.3 → 18.2	B → B	232
EB	297	0	297	0.37 → 0.37	17.6 → 17.6	B → B	202
WB	248	0	248	0.31 → 0.31	16.8 → 16.8	B → B	163

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	58	52%
SB Through	53	48%

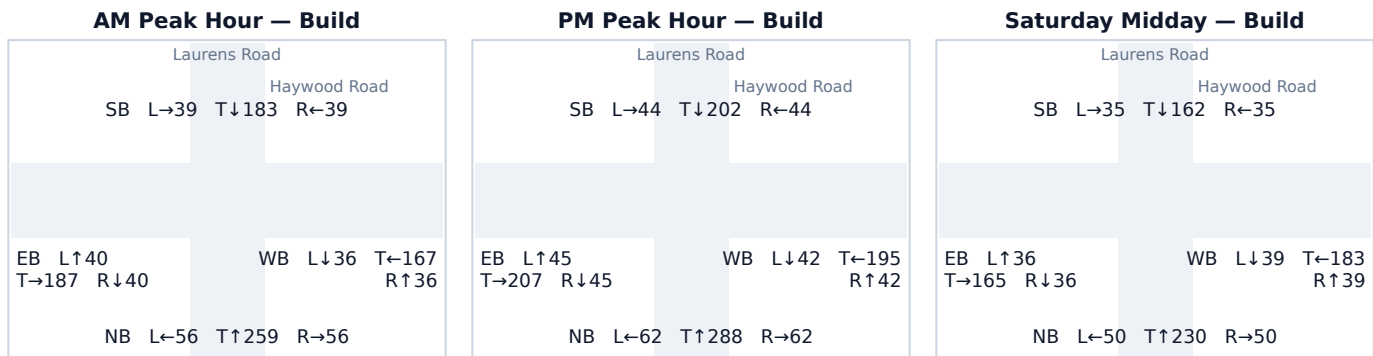
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Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Laurens Road & Haywood Road

Signal ID	greenville-spartanburg-1150
Location	NW Greenville-Spartanburg (SC) · 0.27 mi from site
Added PM peak trips	53
Existing / No-Build (PM)	LOS C · 24.2 s/veh · v/c 0.68
Build (PM)	LOS C · 25.2 s/veh · v/c 0.71
Δ control delay	+1.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	412	0	412	0.51 → 0.51	19.9 → 19.9	B → B	303
SB	290	0	290	0.36 → 0.36	17.5 → 17.5	B → B	196
EB	297	0	297	0.37 → 0.37	17.6 → 17.6	B → B	201
WB	226	53	279	0.28 → 0.34	16.4 → 17.3	B → B	187

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	53	100%

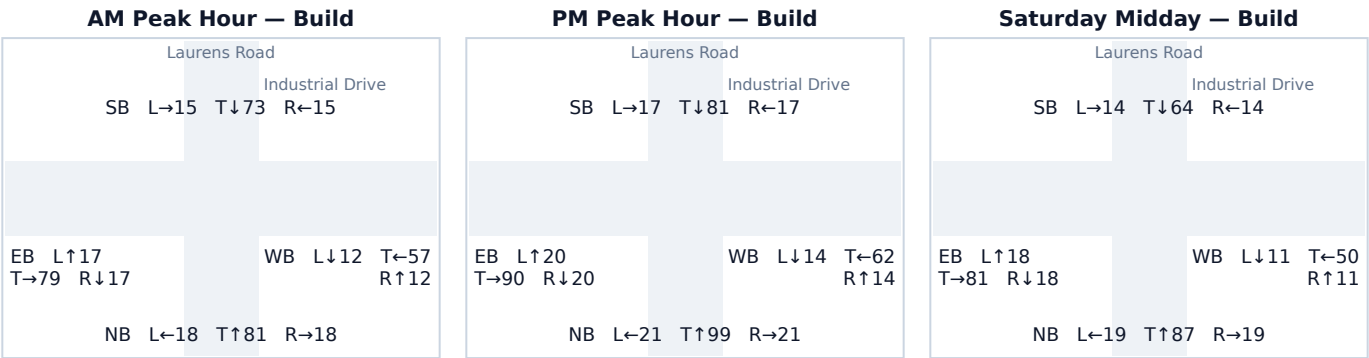
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Laurens Road & Industrial Drive

Signal ID	greenville-spartanburg-166
Location	NW Greenville-Spartanburg (SC) · 0.41 mi from site
Added PM peak trips	38
Existing / No-Build (PM)	LOS B · 16.0 s/veh · v/c 0.24
Build (PM)	LOS B · 16.2 s/veh · v/c 0.26
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	122	20	141	0.15 → 0.17	15.0 → 15.2	B → B	87
SB	115	0	115	0.14 → 0.14	14.9 → 14.9	B → B	69
EB	112	18	130	0.14 → 0.16	14.9 → 15.1	B → B	80
WB	90	0	90	0.11 → 0.11	14.6 → 14.6	B → B	54

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Left	20	53%
EB Right	18	47%

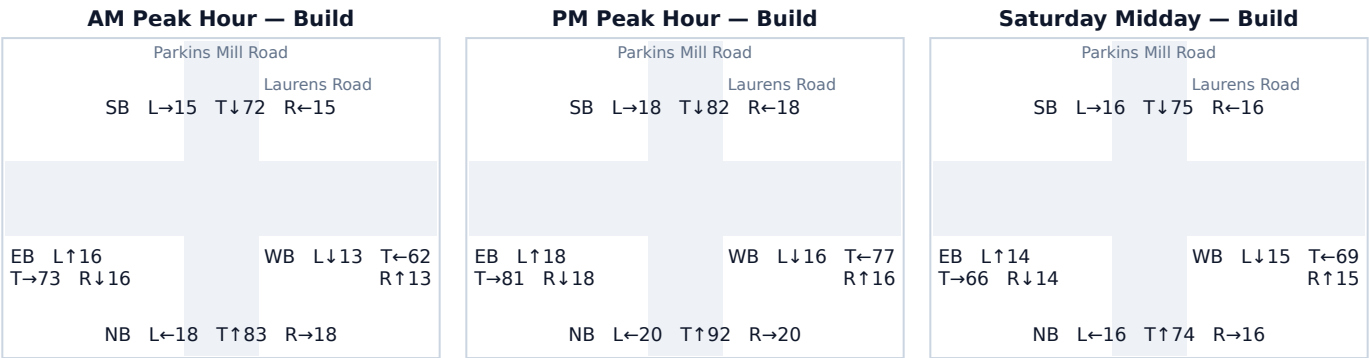
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Parkins Mill Road & Laurens Road

Signal ID	greenville-spartanburg-466
Location	NW Greenville-Spartanburg (SC) · 0.69 mi from site
Added PM peak trips	38
Existing / No-Build (PM)	LOS B · 16.0 s/veh · v/c 0.24
Build (PM)	LOS B · 16.2 s/veh · v/c 0.26
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	132	0	132	0.16 → 0.16	15.1 → 15.1	B → B	81
SB	100	18	118	0.12 → 0.15	14.7 → 14.9	B → B	72
EB	117	0	117	0.14 → 0.14	14.9 → 14.9	B → B	71
WB	90	20	109	0.11 → 0.13	14.6 → 14.8	B → B	66

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	20	53%
SB Through	18	47%

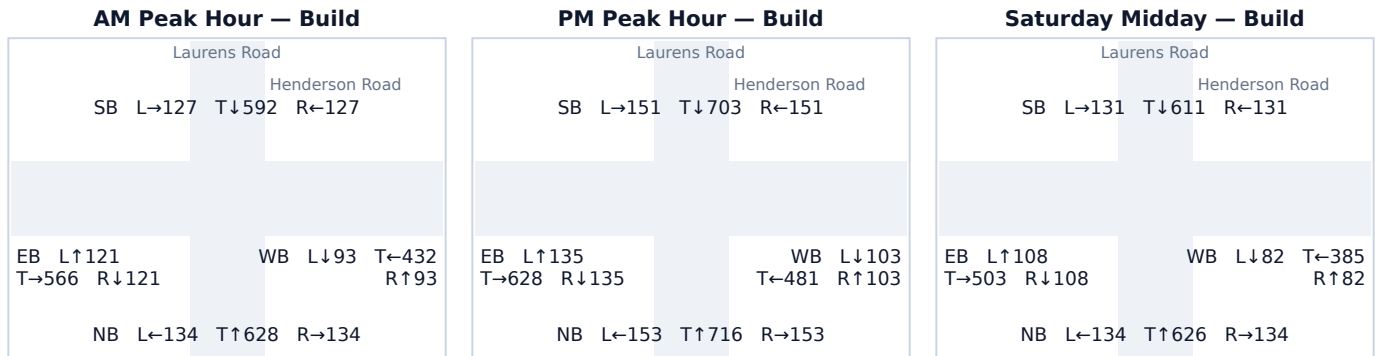
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Laurens Road & Henderson Road

Signal ID	greenville-spartanburg-185
Location	NW Greenville-Spartanburg (SC) · 0.47 mi from site
Added PM peak trips	223
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.88
Build (PM)	LOS F · 300.0 s/veh · v/c 2.01
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	916	107	1,022	1.13 → 1.26	98.4 → 152.4	F → F	1,046
SB	889	116	1,005	1.10 → 1.24	86.2 → 143.1	F → F	1,028
EB	898	0	898	1.11 → 1.11	90.4 → 90.4	F → F	919
WB	687	0	687	0.85 → 0.85	32.7 → 32.7	C → C	629

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	116	52%
NB Through	107	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.